Maybe, he speculated, a more robust theory, which embraced both special relativity and gravity, would come to a different conclusion regarding Mach's ideas. Maybe, he surmised, a generalization of special relativity that incorporated gravity would show that matter, both near and far, determines the force we feel when we accelerate.

Einstein also had a second, somewhat more pressing, reason for turning his attention to gravity. He realized that special relativity, with its central dictum that the speed of light is the fastest that anything or any disturbance can travel, was in direct conflict with Newton's universal law of gravity, the monumental achievement that had for over two hundred years predicted with fantastic precision the motion of the moon, the planets, comets, and all things tossed skyward. The experimental success of Newton's law notwithstanding, Einstein realized that according to Newton, gravity exerts its influence from place to place, from the sun to the earth, from the earth to the moon, from any-here to any-there, instantaneously, in no time at all, *much faster than light.* And that directly contradicted special relativity.

To illustrate the contradiction, imagine you've had a really disappointing evening (hometown ball club lost, no one remembered your birthday, someone ate the last chunk of Velveeta) and need a little time alone, so you take the family skiff out for some relaxing midnight boating. With the moon overhead, the water is at high tide (it's the moon's gravity pulling up on bodies of water that creates the tides), and beautiful moonlight reflections dance on its waving surface. But then, as if your night hadn't already been irritating enough, hostile aliens zap the moon and beam it clear across to the other side of the galaxy. Now, certainly, the moon's sudden disappearance would be odd, but if Newton's law of gravity was right, the episode would demonstrate something odder still. Newton's law predicts that the water would start to recede from high tide, because of the loss of the moon's gravitational pull, about a second and a half *before* you saw the moon disappear from the sky. *Like a sprinter jumping the gun, the water would seem to retreat a second and a half too soon.*

The reason is that, according to Newton, at the very moment the moon disappears its gravitational pull would *instantaneously* disappear too, and without the moon's gravity, the tides would immediately start to